

# 电控环境配置指南

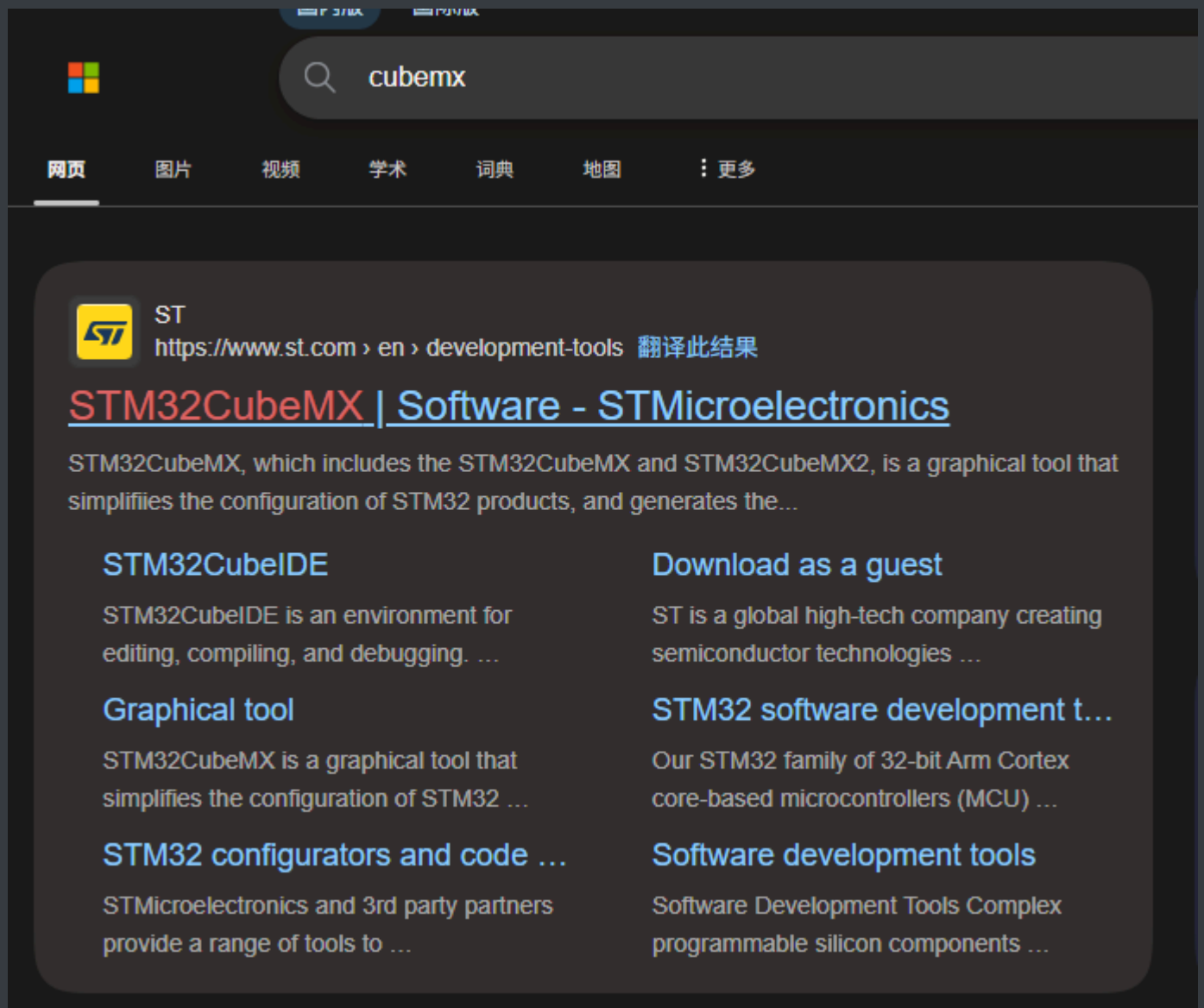
本文档旨在为准备进入电控的小登提供环境配置指南

author: rediduck, lary

## CubeMX安装指南

CubeMX是使用stm32开发不可避免的一个工具，不管使用什么工作流都需要下载

1、搜索官网并注册个人账号（可能需要科学上网）



2、往下翻知道你看到了这个界面

该网站提供您所使用的语言版本

跳转本地站点

不用，谢谢

三

ST



Q Search...

Pinout Configuration  
Clock tree initialization  
Peripherals configuration  
Middleware addition and configuration  
Code and project generation  
Available for all STM32 MCUs and MPUs

STM32CubeMX

High Performance MCUs

F2 F4 H5 F7 H7 H7R/S N6

Mainstream MCUs

G0 C0 F0 F1 F3 G4

Ultra-low-power MCUs

L0 U0 L1 L4 L4+ L5 U3 U5

Wireless MCUs

WB0 WL WL3 WB WBA

MPU

MP1 MP2

STM32CubeMX2

Mainstream MCUs

C5

Get Software

| Software name | Supported hardware                | ECCN                        | OS      | Version     | Action                                                                                         |
|---------------|-----------------------------------|-----------------------------|---------|-------------|------------------------------------------------------------------------------------------------|
| STM32CubeMX   | All MCU/MPU except STM32C5 Series | NEC (EU)<br>5D992.c<br>(US) | Windows | 6.17.0 Late |  Download |
| STM32CubeMX2  | STM32C5 Series                    | NEC (EU)<br>5D992.c<br>(US) | Windows | 1.0.0 Late  |  Download |

STMicroelectronics recommends always keeping your software up to date

Featured Products

ST67W coprocessor series

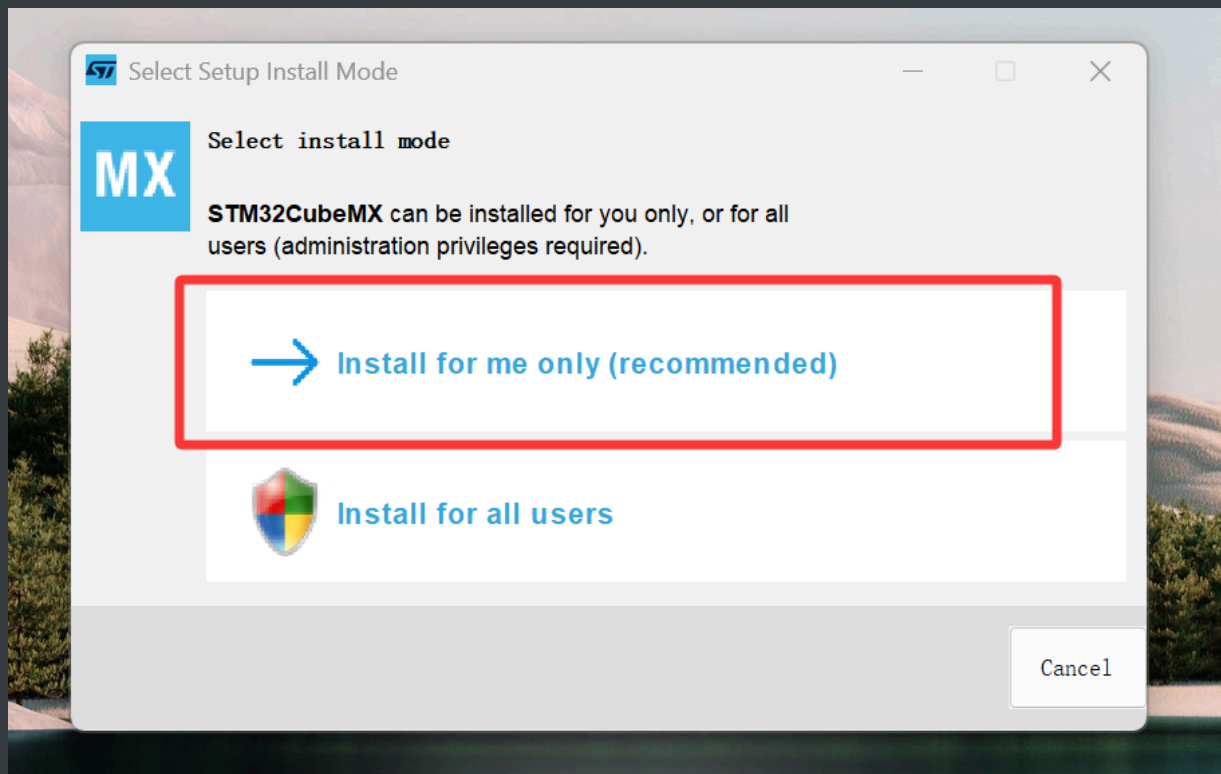
The first Wi-Fi coprocessor series

STM32N6 series

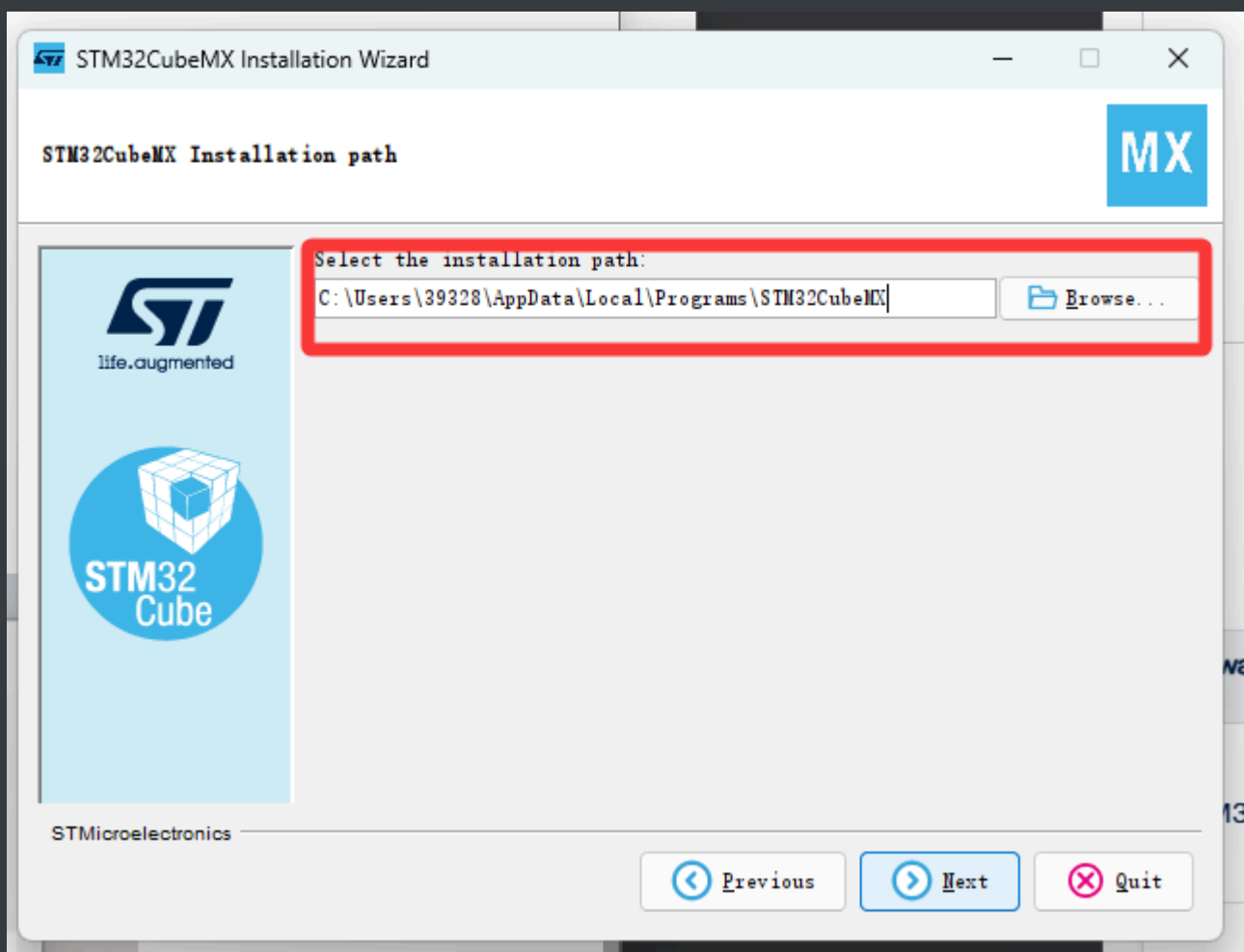
NPU-accelerated, Cortex-M55 STM32 microcontrollers

点击下载STM32CubeMX，不要下成2了（因为是才出来的，队里的老登还没有研究过）

3、点击安装程序，跟着图片一步一步来

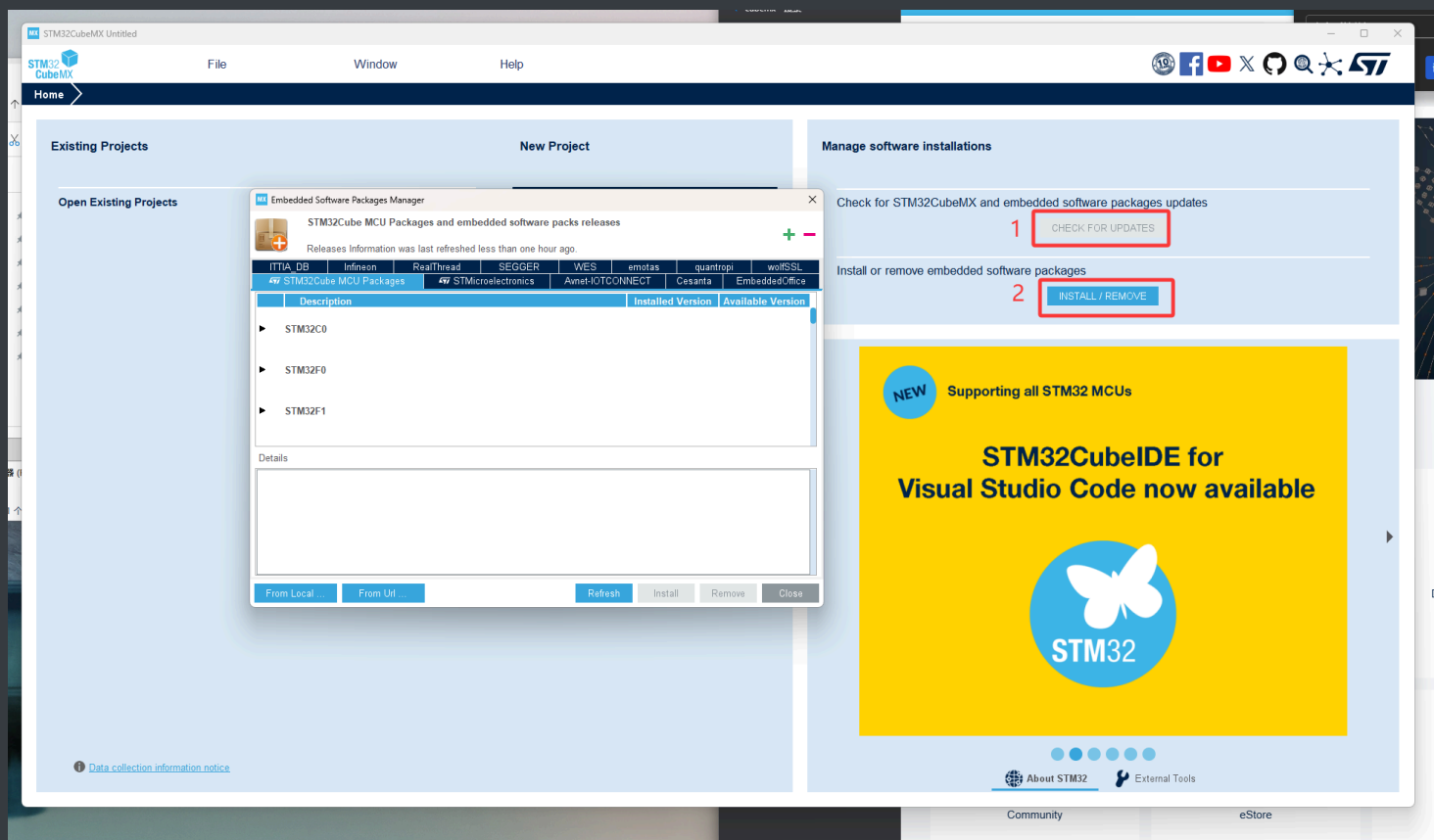


一路Next, 直到这个界面



选一个你能找到的路径安装, 可以不放在C盘。(然后就是静静等待下载)

4、接着你将会在桌面看到相应软件，点开它，检查软件更新并下载相应芯片驱动包



分别选取STM32F1和STM32F4系列最新的驱动包，点击下载



## STM32Cube MCU Packages and embedded software packs releases



Releases Information was last refreshed less than one hour ago.

|                        |                    |                  |         |                |        |           |         |
|------------------------|--------------------|------------------|---------|----------------|--------|-----------|---------|
| ITIA_DB                | Infineon           | RealThread       | SEGGER  | WES            | emotas | quantropi | wolfSSL |
| STM32Cube MCU Packages | STMicroelectronics | Avnet-IOTCONNECT | Cesanta | EmbeddedOffice |        |           |         |

| Description                                                                                    | Installed Version | Available Version |
|------------------------------------------------------------------------------------------------|-------------------|-------------------|
| STM32F0                                                                                        |                   |                   |
| STM32F1                                                                                        |                   |                   |
| STM32F2                                                                                        |                   |                   |
| STM32F3                                                                                        |                   |                   |
| STM32F4                                                                                        |                   |                   |
| <input checked="" type="checkbox"/> STM32Cube MCU Package for STM32F4 Series (Size : 925.0 MB) |                   | 1.28.3            |
| <input type="checkbox"/> STM32Cube MCU Package for STM32F4 Series (Size : 925.0 MB)            |                   | 1.28.2            |
| <input type="checkbox"/> STM32Cube MCU Package for STM32F4 Series (Size : 895.0 MB)            |                   | 1.28.1            |

## Details

From Local ...

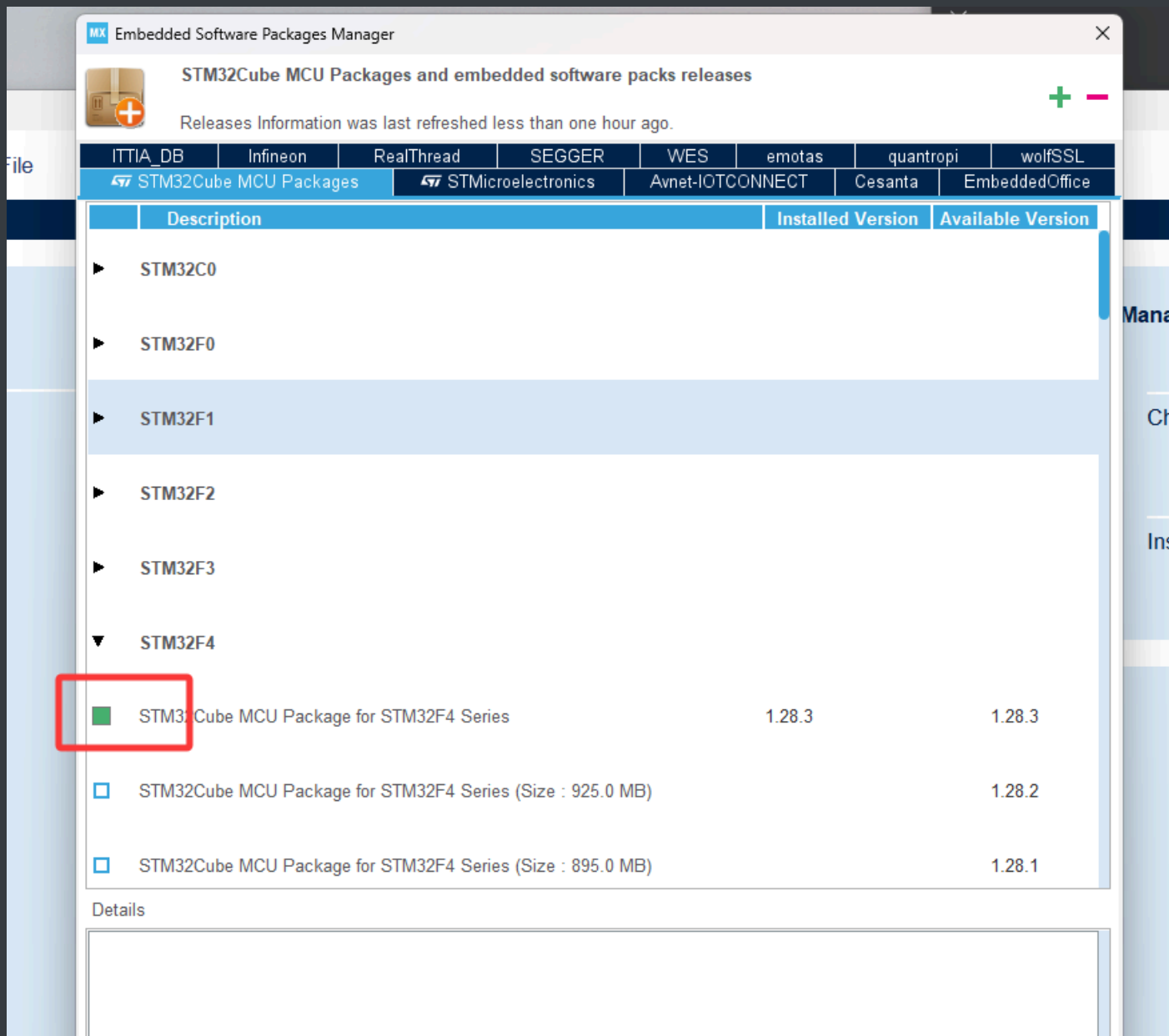
From Url ...

Refresh

Install

Remove

Close



成功安装驱动包后会显示绿豆，后续使用cubemx时它可能会更新驱动包或者软件，下载保持在最新版本即可

安装完成后请看CubeMX使用教程

## OZONE+JLINK安装指南

- ozone各个版本都支持jlink，推荐Ozone3.24 32-bit和J-Link7.52，这一套可以支持jlink和dap-link的调试（包括ATK无线调试器）
- 先安装Ozone，再安装Jlink

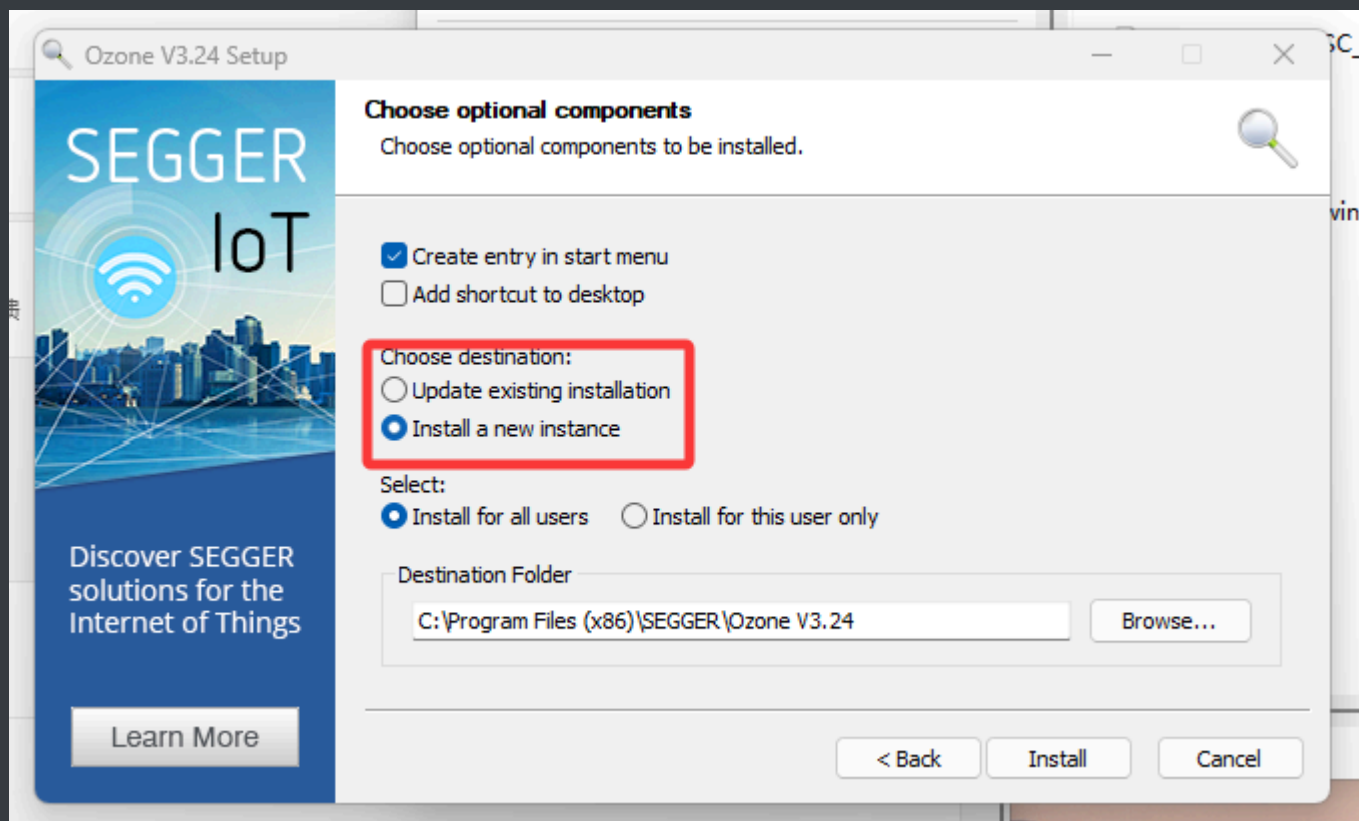
1、搜索网址 <https://www.segger.com/downloads/jlink/>，安装3.24版本

# Ozone - The J-Link Debugger



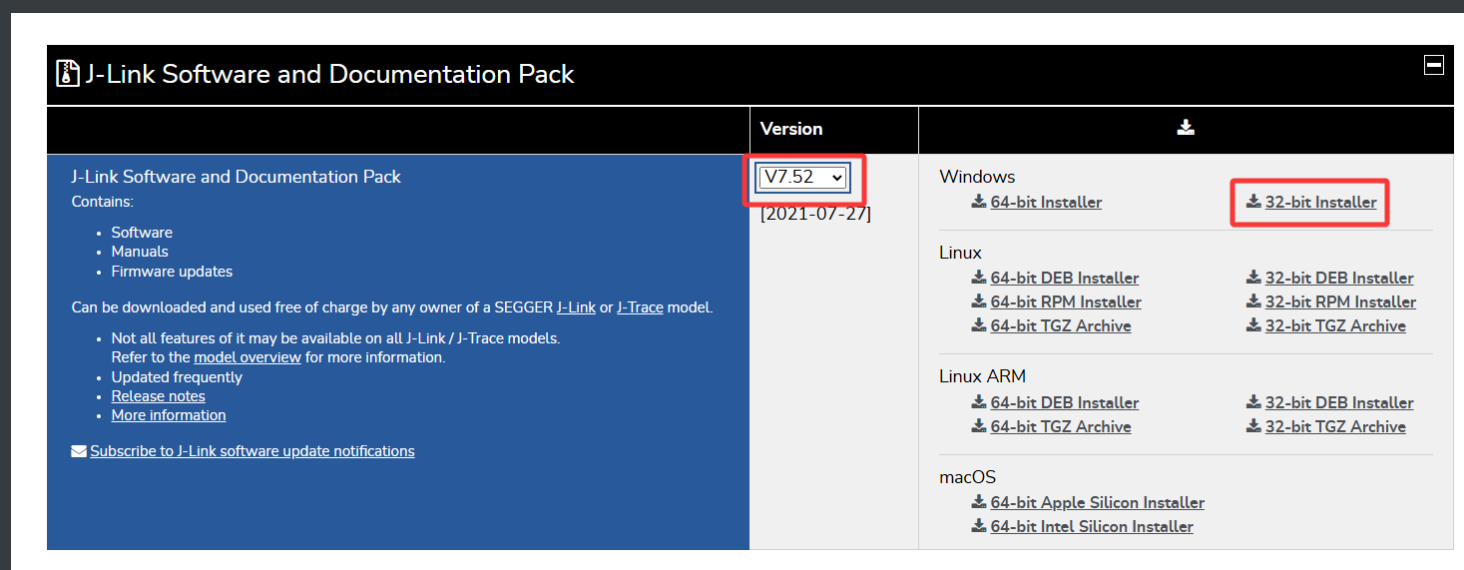
|                                                                                                                                                                                                                                                                                                                                                         | Version                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>Ozone - The J-Link Debugger</div> <ul style="list-style-type: none"><li>• Full-featured graphical debugger for embedded applications.</li><li>• Debug any embedded application on C source and assembly level.</li><li>• Load applications built with any toolchain / IDE or debug the target's resident application without any source.</li></ul> | <div>V3.24 ▾</div> <div>[2021-06-17]</div> | <div>Windows</div> <div> <a href="#">64-bit Installer</a>  <a href="#">32-bit Installer</a></div> <div>Linux</div> <div> <a href="#">64-bit DEB Installer</a>  <a href="#">32-bit DEB Installer</a></div> <div> <a href="#">64-bit RPM Installer</a>  <a href="#">32-bit RPM Installer</a></div> <div> <a href="#">64-bit TGZ Archive</a>  <a href="#">32-bit TGZ Archive</a></div> <div>macOS</div> <div> <a href="#">64-bit Intel Silicon Installer</a></div> |

2、进入安装程序，next几次直到



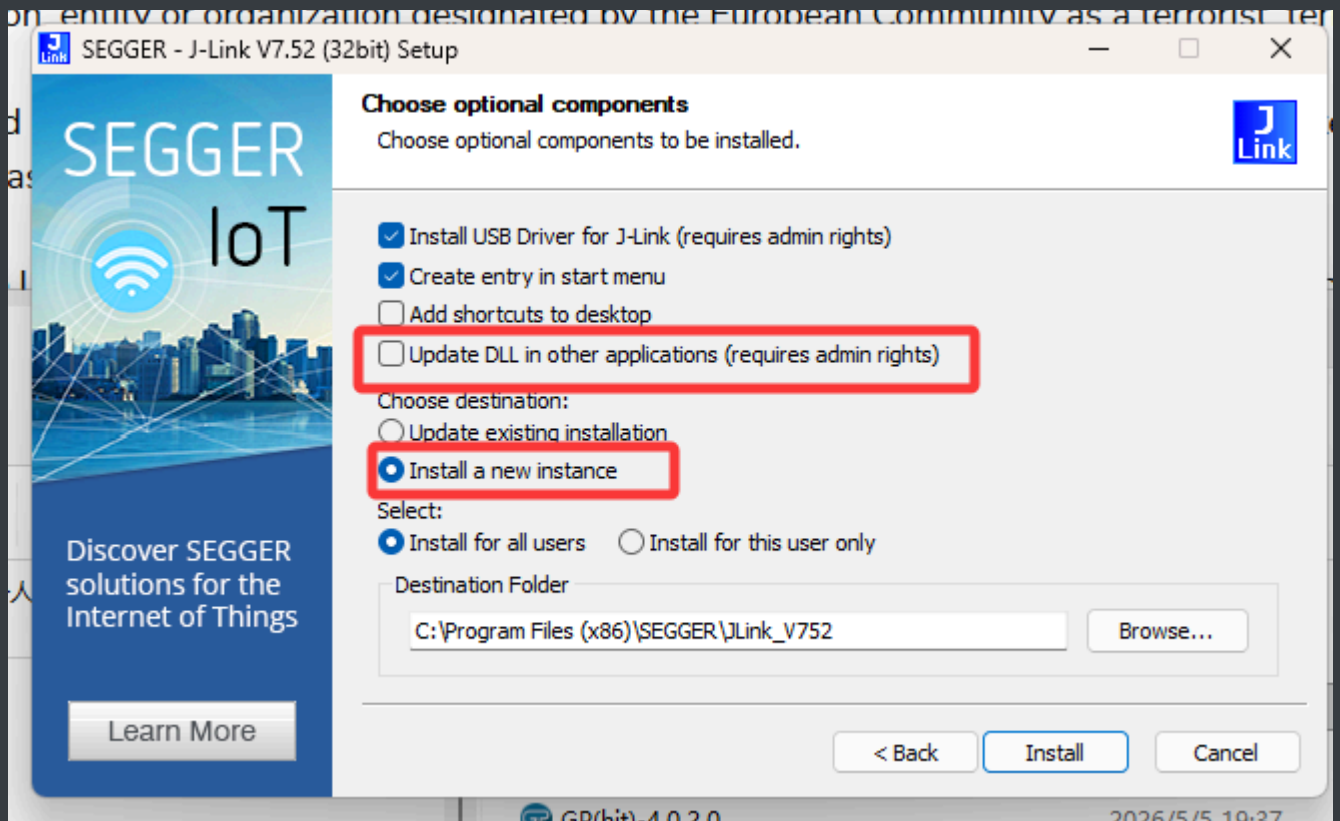
选择安装一个新实例，后续一路确认即可

### 3、回到官网，下载JLink驱动



### 4、打开安装程序





这一步注意不要勾选date dll in other application，否则jlink会把ozone里面老的驱动和启动项替代掉。choose destination和ozone一样，选择install a new instance。如果安装了老的相同版本的jlink，请先卸载（版本相同不用管，直接新装一个）

5、在开始菜单中搜索dll，选择这个程序运行



全部

应用

文档

网页

设置

文件夹

照片



200

### 最佳匹配



J-Link DLL Updater V7.52

应用

商店 2



DLL系统修复



电脑异常修复 - DLL & DirectX修复工具

搜索网页 11+



dll - 查看更多搜索结果



dll修复工具



dll修复工具.site



dll修复工具免费



大连理工大学

设置 1



@%SystemRoot%\System32\WorkFolder...



J-Link DLL Updater V7.52

应用



打开



以管理员身份运行



打开文件位置



固定到“开始”屏幕



固定到任务栏

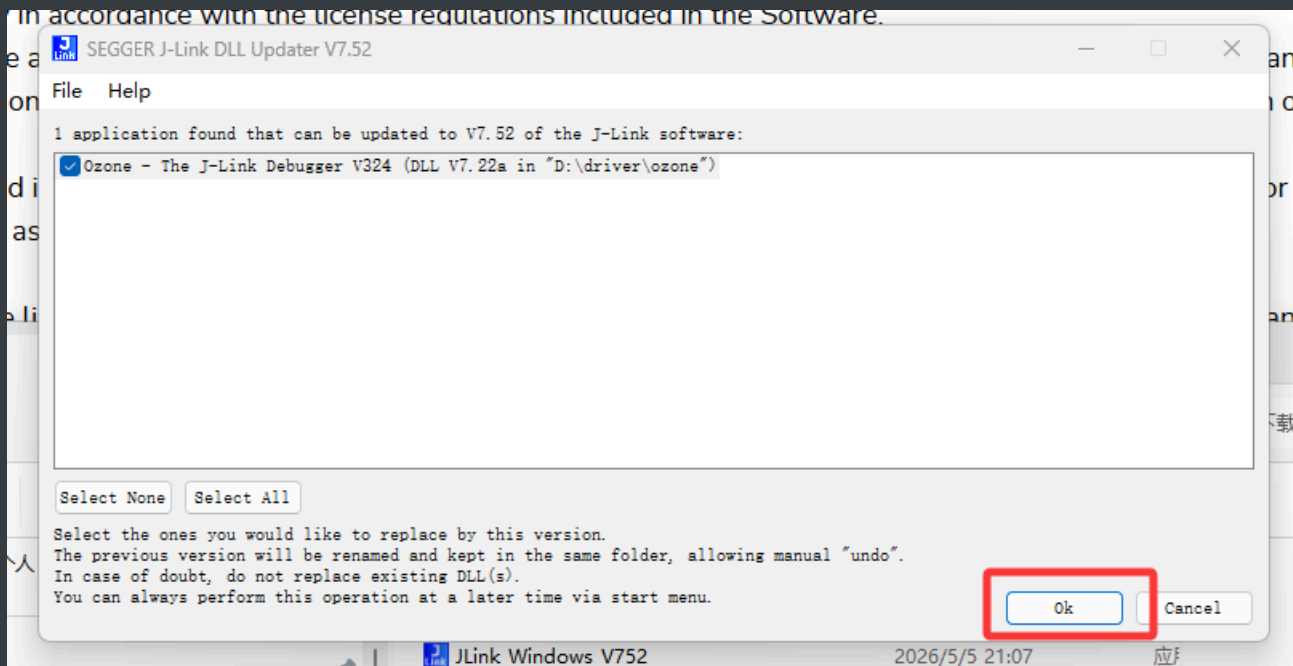


卸载



dll





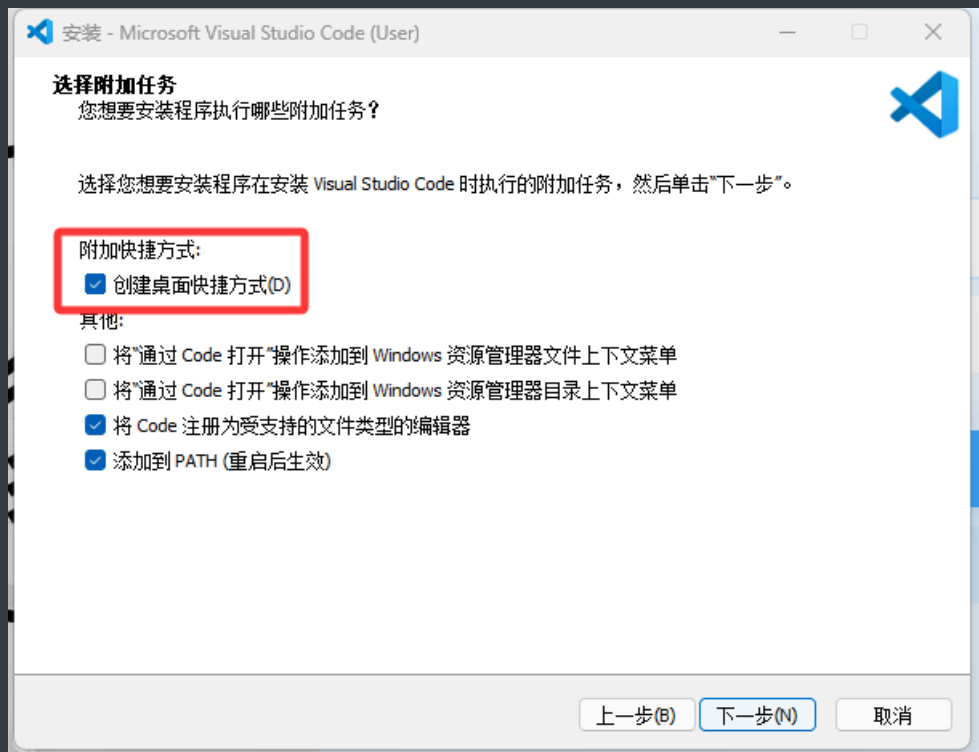
这样就可以使用新版的调试器和老版的驱动

安装完成后请查看ozone使用教程

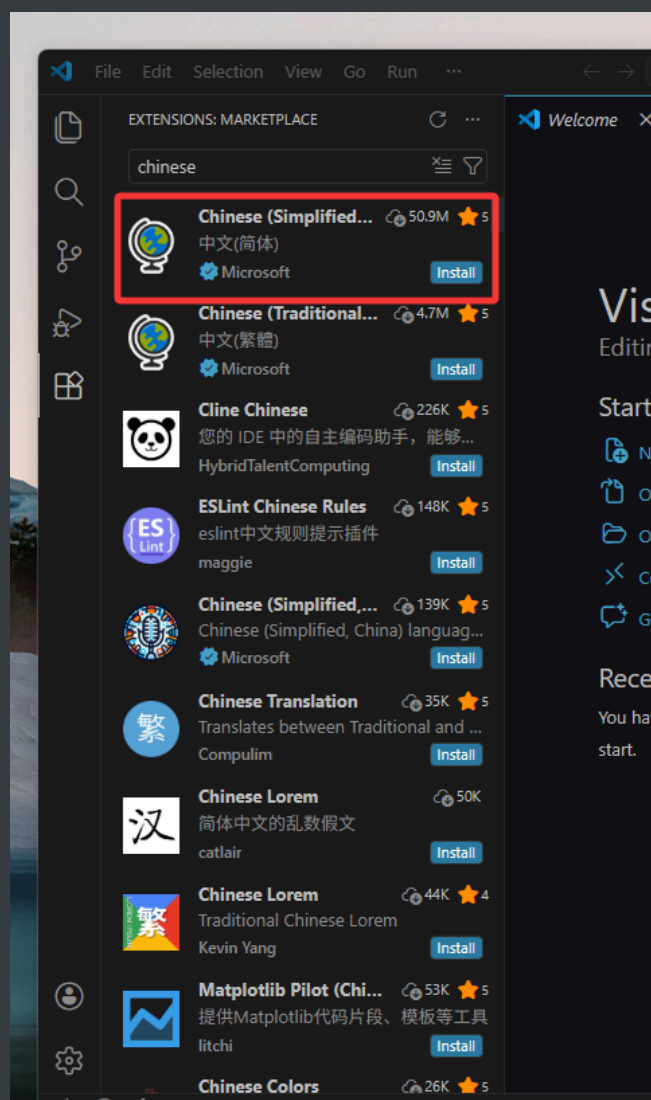
## vscode+cmake+stm32cube插件安装

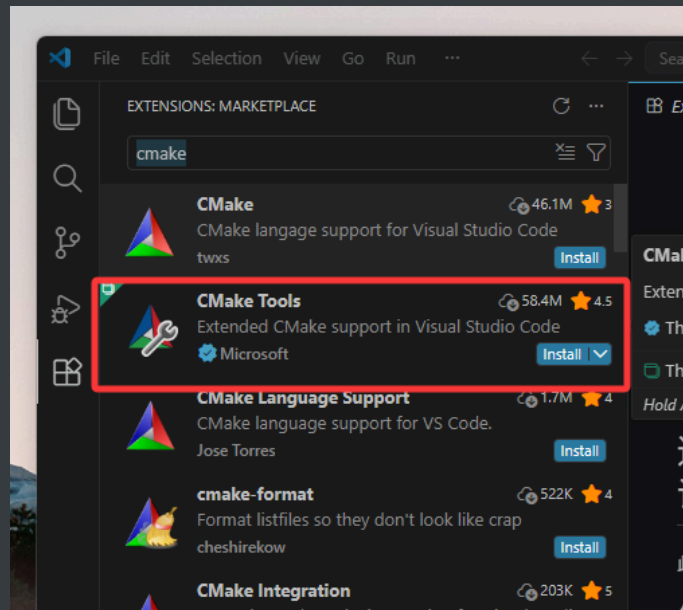
目前战队正在使用的工作流是基于cmake的，可爱的cube为我们准备了插件可以帮我们安装编译链之类的

1、前往vscode官网下载软件（安装时注意选择创建桌面快捷方式）



2、进入vscode（建议用github账号登录），安装一些插件





配置完成，后续可看 workflow 使用说明

## stm32tool+cpkg 安装

由于目前队里的包比较多，因此有必要引入包管理以及便于初始化项目的 stm32tool

- 1、进入战队 github 仓库 ([HITSZ-WTRobot/stm32tool: Command line tool for initializing STM32 + Git projects](https://github.com/HITSZ-WTRobot/stm32tool))

https://github.com/HITSZ-WTRobot/stm32tool

🏠

🔍

🌐

🌟

🔍

stm32toolPublic

Edit Pins

Watch0

Fork0

Star1

main

Go to file

+

Code

👤 syhanjin

chore(codex): add stm32tool skill

4885213 · last week

|                         |                                        |              |
|-------------------------|----------------------------------------|--------------|
| 📁 .github/workflows     | feat: add CI workflows for cross-pl... | 5 months ago |
| 📁 .idea                 | shit: 不想写提交消息了                         | 8 months ago |
| 📁 skills/stm32tool      | chore(codex): add stm32tool skill      | last week    |
| 📁 src                   | feat(logging): stream verbose co...    | 3 weeks ago  |
| 📄 .gitignore            | chore(codex): add stm32tool skill      | last week    |
| 📄 AGENTS.md             | fix(windows): use STM32CubeMX_...      | 3 weeks ago  |
| 📄 Cargo.lock            | chore(release): bump version to 0...   | 3 weeks ago  |
| 📄 Cargo.toml            | chore(release): bump version to 0...   | 3 weeks ago  |
| 📄 LICENSE               | Initial commit                         | 8 months ago |
| 📄 README.md             | feat(logging): add verbose debug ...   | 3 weeks ago  |
| 📄 convert.py            | feat(windows): add installer scripts   | 3 weeks ago  |
| 📄 install-stm32tool.cmd | feat(windows): add installer scripts   | 3 weeks ago  |
| 📄 install-stm32tool.ps1 | fix(windows): use STM32CubeMX ...      | 3 weeks ago  |

About

Command line tool for initializing STM32 + Git projects

📖 Readme

📄 MIT license

📈 Activity

📋 Custom properties

🌟 1 star

👁 0 watching

🔗 0 forks

📋 Audit log

Report repository

Releases9

📦 v0.3.1

Latest

3 weeks ago

+ 8 releases

Packages

No packages published

点击release下载

v0.3.1

Latest

github-actions

released this 3 weeks ago · 1 commit to main since this release

v0.3.1

3b64ba5

Full Changelog: [v0.3.0...v0.3.1](#)

Assets

4

stm32tool-linux-v0.3.1

4.49 MB

3 weeks ago

stm32tool-windows-v0.3.1.exe

3.08 MB

3 weeks ago

Source code (zip)

3 weeks ago

Source code (tar.gz)

3 weeks ago

下载完成后解压，运行自动脚本

| 名称                | 修改日期           | 类型                    |
|-------------------|----------------|-----------------------|
| .github           | 2026/5/5 22:54 | 文件夹                   |
| .idea             | 2026/5/5 22:54 | 文件夹                   |
| src               | 2026/5/5 22:54 | 文件夹                   |
| .gitignore        | 2026/5/5 22:54 | Git Ignore 源文件        |
| AGENTS            | 2026/5/5 22:54 | Markdown File         |
| Cargo.lock        | 2026/5/5 22:54 | LOCK 文件               |
| Cargo             | 2026/5/5 22:54 | Toml 源文件              |
| convert           | 2026/5/5 22:54 | Python 源文件            |
| install-stm32tool | 2026/5/5 22:54 | Windows 命令脚本          |
| install-stm32tool | 2026/5/5 22:54 | Windows PowerShell 脚本 |
| LICENSE           | 2026/5/5 22:54 | 文件                    |
| README            | 2026/5/5 22:54 | Markdown File         |

运行过程中需要手动选择一下安装路径，选后即安装完成

2、进入战队驱动库 ([HITSZ-WTRobot-Packages/cpkg](#): 与驱动库搭配使用的一套包管理器)

cpkgPublic

Edit Pins

Watch0

Fork0

Stars

main

Go to file

+

Code

syhanjin docs(skill): add cpkg agent skill ✓07bd86b · last week

|                   |                                         |             |
|-------------------|-----------------------------------------|-------------|
| .github/workflows | ci(release): add macos and arm64 ...    | 2 weeks ago |
| skills/cpkg       | docs(skill): add cpkg agent skill       | last week   |
| src               | feat(cli): add colored package listi... | 2 weeks ago |
| .codex            | chore(release): bump version to 0...    | 3 weeks ago |
| .gitignore        | chore(repo): add local index and i...   | 3 weeks ago |
| AGENTS.md         | docs(skill): add cpkg agent skill       | last week   |
| Cargo.lock        | ci(release): add macos and arm64 ...    | 2 weeks ago |
| Cargo.toml        | ci(release): add macos and arm64 ...    | 2 weeks ago |
| LICENSE           | Initial commit                          | 3 weeks ago |
| README.md         | feat(cli): add colored package listi... | 2 weeks ago |

README

GPL-3.0 license

About

与驱动库搭配使用的一器

Readme

GPL-3.0 license

Activity

Custom properties

0 stars

0 watching

0 forks

Audit log

Report repository

Releases12

v0.3.2Latest2 weeks ago

+ 11 releases

Packages

| Assets8                                      |         |     |
|----------------------------------------------|---------|-----|
| cpkg-v0.3.2-aarch64-apple-darwin.tar.gz      | 1.37 MB | 2 w |
| cpkg-v0.3.2-aarch64-pc-windows-msvc.zip      | 1.17 MB | 2 w |
| cpkg-v0.3.2-aarch64-unknown-linux-gnu.tar.gz | 1.45 MB | 2 w |
| cpkg-v0.3.2-x86_64-apple-darwin.tar.gz       | 1.43 MB | 2 w |
| cpkg-v0.3.2-x86_64-pc-windows-msvc.zip       | 1.23 MB | 2 w |
| cpkg-v0.3.2-x86_64-unknown-linux-gnu.tar.gz  | 1.5 MB  | 2 w |
| Source code (zip)                            |         | 2 w |
| Source code (tar.gz)                         |         | 2 w |

下载后解压，修改用户的环境变量



